Submission Note

ILS Venture Progress & Readiness Workflow

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Stack: FastAPI, SQLite, offline React, no Vite or npm registry.
Effort: 5 hours.
Live demo: https://ils-venture-desk.vercel.app/
Source: https://github.com/iamsankeerth/ils-venture-desk
Demo video: https://www.loom.com/share/3da814ebe2bb4620a10dc82c2759641e

1. Assumptions

- Tracker sector `AI/Product` is a legacy label, not a business model. Criteria attach to `Consumer`, `Services`, `AI_B2C` or `AI_B2B`.
- Cases A and B have unresolved governing models. The prototype preserves both tracks and shows the tracker-assigned line as assigned, never as confirmed.
- Business lines are the reconciliation unit. Case B Services and SaaS evidence never mix. Case C's historical total is isolated from the three mock records.
- Missing source data stays `Unknown`. It is never coerced to zero, false, or a failed criterion.
- The Programme Reviewer role assumption is used only for this exercise. `X-ILS-Role` is a simulated seam, not production authentication.
- Amounts are stored and transported as integer paise. Rupees are display-only.
- All names and evidence are anonymised or synthetic. No external accounts, students, mentors, or live messages were contacted.

2. Workflow and Data Model

The flow is import or seed a venture with original status and provenance, submit a short update against a criterion and business line, review it, recompute requirements, and record advancement or direct entry. The update captures metric, period, evidence reference, blocker and next action.

How updates are worked. A Student Founder opens a venture and selects **Requirements → Submit Update**, or uses the workspace Submit Update button. The founder chooses the governing line and criterion, enters the metric, count or amount and verbatim period, attaches or creates the evidence record, and records a blocker and next action. The update and evidence are saved as `UNREVIEWED`; they are claims, not verified proof. A Programme Reviewer opens **Work Queue → Review Evidence**, inspects the submitted measurements, records a reason, and approves or requests changes. Approval activates the observation and recomputes the criterion; requested changes do not count. A Programme Owner configures unresolved programme rules, a Programme Reviewer records transitions, and the Work Queue surfaces pending evidence, clarifications and follow-ups.

SQLite holds nine tables: ventures, business lines, a 38-row criteria catalogue, progress updates, evidence records, period-level observations, review history, transition decisions, and MSL eligibility decisions. Two independent concepts drive the model:

- `evidence_status` records whether the document is `UNREVIEWED`, `APPROVED`, or `CHANGES_REQUESTED`.
- `criterion_result` records whether approved evidence satisfies the criterion: `UNKNOWN`, `IN_PROGRESS`, `MET`, `NOT_MET`, or `BLOCKED`.

Seven venture state dimensions are stored independently: development stage, programme level, MSL eligibility, advancement history, fundraising preference, MSL handover, and MSL acceptance. Transitions are immutable audit rows with a criteria snapshot. MSL eligibility is never inferred from fundraising preference, handover, or acceptance. New MSL certification requires a confirmed model and every required MSL criterion `MET`; imported eligibility is preserved with provenance.

Financial claims are reconciled per business line. Case B shows INR 100,000 billed and INR 60,000 claimed/submitted, with a INR 40,000 arithmetic difference and no verified outstanding balance. Case C keeps the claimed INR 80,000 September earnings and INR 70,000 payout as separate observations, surfacing a claimed INR 10,000 unreconciled difference. The verified row remains `Unknown` until the evidence is approved.

3. Final Safeguards

The final pass hardened two workflow rules without changing the supplied cases. A direct-entry target below the current level is rejected before any transition is written. Invalid programme-rule configurations are rejected instead of being silently accepted, and unconfigured rules remain `BLOCKED`. Historical direct entry, review history and the original cases remain intact.

4. Unresolved Rules

Case A's model is unresolved because tracker usage criteria are consumer-style while recent activity targets agencies. The prototype shows `AI B2C (assigned in tracker)` and `AI B2B (candidate, agency motion)` side by side and blocks model-dependent advancement until a Programme Owner clarifies. Case B likewise requires Services and SaaS clarification.

Eight threshold properties remain open in the guideline ranges: MSL Services 10 to 20 customers and 30 to 40 percent repeat/retention; MSL AI B2B 1 to 2 customers; INR 2 lakh or more revenue basis; INR 125,000 cumulative basis and period; 35 WAU over four weeks aggregation; pilot qualification; and organiser retention definition. They are flagged for configuration. The five L0 qualitative items have no supplied evidence and remain `Unknown`; completing a demand bar never completes Level 0.

5. Persistence, Mocks and Dependencies

Local SQLite persistence is real. Updates, review decisions, transitions, clarifications, and MSL decisions survive restart and appear in export. The seed, evidence excerpts, review history, and programme decisions are mock exercise data. No document storage, OCR, email, WhatsApp, scheduling, or third-party integration is claimed. Export is a full JSON dump; import round-trips all nine tables.

The application runs locally with the persistent `backend/ils.db` database. A public Vercel demo is available at the link above; it sets `ILS_DB=/tmp/ils.db`, so data written in the hosted demo is ephemeral and resets when the Vercel function instance is recycled. The local app remains the persistent version.

No purchase or paid service was used. Runtime dependencies are FastAPI, Pydantic, Uvicorn, standard-library SQLite, and the vendored React/Tailwind/Lucide frontend; tests use pytest and Playwright. Vercel hosts the short demo with ephemeral `/tmp` storage. Production would add a managed database, object storage, Mesa SSO, backups, monitoring and audit retention, which are out of scope.

6. Effort and AI Assistance

Total effort is 5 hours across planning, domain modelling, implementation, adversarial audit, remediation, browser verification, final corrections, packaging and deployment preparation.

Material AI assistance: an opencode coding agent for implementation, refactoring, test authoring, and Playwright QA scripts. Business rules came from the supplied assignment. Review decisions in the product remain human decisions.

7. Access Control, Mentor Work and MSL Handover

Production access control belongs in Mesa identity and server-side programme-role membership, enforced on every route and mirrored in the UI capability matrix. Student Founders submit updates but cannot review evidence, transition, clarify models, certify MSL, import, export, or reset. Programme Reviewers verify evidence and approve transitions. Programme Owners clarify models and configure rules. Mentors record outcomes but do not approve evidence.

Mentor pre-work fits before a founder submits. Mentor post-work fits after review, when the reviewer's decision and revised blockers become follow-up items. The prototype exposes both through the staff queue and mentor outcome records; scheduling and messaging are omitted.

MSL handover fits after eligibility and before acceptance. The handover preview shows eligibility provenance, open MSL criteria, financial reconciliation, and missing items. Handover and acceptance stay `Unknown` until separately recorded; eligibility never infers either.

8. Known Limitations

- `X-ILS-Role` is a demo seam, not authentication.
- The offline React compatibility runtime is not a production React replacement.
- No draft-configuration workflow exists; invalid active configurations are rejected.
- Same-level direct entry remains allowed. The correction covers only downward direct entry. No separate downgrade workflow was added.

9. Paid-Pilot Priorities

1. Replace the role header with Mesa SSO and server-side programme roles.
2. Fix the eight open thresholds with the programme owner and version every configuration change.
3. Add evidence file storage, review retention, and export retention.
4. Instrument time-to-review, evidence rejection rate, and stale follow-ups.